22/01/2025

Code-C Phase-3

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122018

MM : 720 NCERT Booster Test Series for NEET (RM)-2024-25_T02C **Time : 200 Min.**

Topics Covered:
Physics: Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves, Electric Charges & Fields, Electrostatic Potential and Capacitance, Current Electricity
Chemistry: Equilibrium, Redox Reactions, Electrochemistry, Solutions, Chemical Kinetics, Organic Chemistry: Some Basic Principle & Techniques, Hydrocarbons, Haloalkanes and Haloarenes
Botany: Anatomy of Flowering Plants, Plant Kingdom, Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plants
Zoology: Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration, Animal Kingdom, Structural Organisation in Animals– Animal Morphology (Cockroach & Frog)

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

SECTION-A

1. Inside a hollow charged spherical conductor, the electric potential

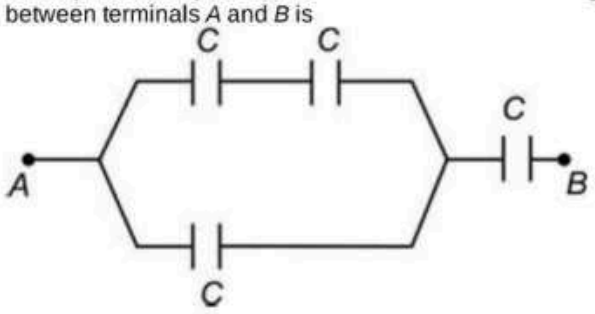
(1) Is constant

(2) Varies directly as the distance from the centre

(3) Varies inversely as the distance from the centre

(4) Varies inversely as the square of the distance from the centre

2. The equivalent capacitance of the circuit shown in the figure between terminals *A* and *B* is



(1) $\frac{C}{5}$

(2) $\frac{2C}{5}$

(3) $\frac{3C}{5}$

(4) $\frac{4C}{5}$

3. **Assertion (A) :** Electric force on the positive charge act in the direction of electric field.
Reason (R) : Electric field intensity vector due to a point charge will be different at different point.
In the light of above statements, select the correct option

(1) Both (A) and (R) are true, (R) correct explanation of (A)

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) (A) is false, (R) is true

4. An electric dipole of moment $\vec{P}$ placed in uniform electric field $\vec{E}$ has minimum potential energy. The work done by external agent to rotate this dipole very slowly upto which it has maximum potential energy.

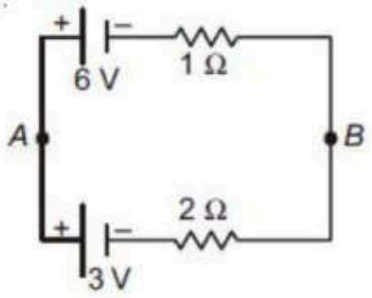
(1) $-PE$

(2) $2PE$

(3) $\frac{PE}{2}$

(4) Zero

5. In the given circuit diagram, the potential difference between *A* and *B* is



(1) 3 V

(2) 5 V

(3) 6 V

(4) 9 V

6. A body takes 4 minutes to cool from 80°C to 60°C. If the temperature of the surrounding is 25°C the time taken by it to cool from 60°C to 40°C will be

(1) 7.2 minutes

(2) 5.2 minutes

(3) 4.9 minutes

(4) 8.1 minutes

7. Efficiency of a Carnot engine working between a source at T_1 K and sink at T_2 K is 0.5. When the temperature of the sink is decreased by 100°C keeping temperature of source constant, its efficiency becomes 0.6. Temperature of source T_1 is

(1) 500 K

(2) 1000 K

(3) 600 K

(4) 400 K

8. Fundamental frequency of open organ pipe of length *L* is *f*. If 20% of its length is dipped in water, then fundamental frequency of remaining length is

(1) $\frac{5}{8}f$

(2) $\frac{4f}{5}$

(3) $\frac{3}{8}f$

(4) $\frac{f}{5}$

9. Breaking stress of wire depends on

(1) Length of wire

(2) Area of cross-section of wire

(3) Nature of material

(4) Volume of wire

10. Initial kinetic energy required to make a body move to infinity from the earth surface is

(1) Infinite

(2) $2mgR$

(3) $\frac{1}{2}mgR$

(4) mgR

11. Distance between plates of a parallel plate charged capacitor (disconnected from battery) is increased. The quantity that remains constant is

(1) Potential difference between plates

(2) Charge

(3) Electric field between plates

(4) Both (2) and (3)

12. Two positive point charges of 9 μC & 6 μC are 10 cm apart. The work done in bringing them to a separation of 4 cm is

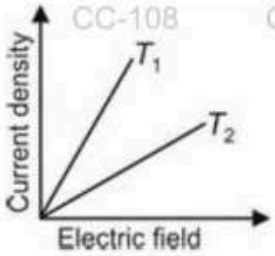
(1) 5.25 J

(2) 6.29 J

(3) 7.29 J

(4) 8.25 J

13. The current density in a metallic conductor is plotted against electric field in a conductor at two different temperatures T_1 & T_2 choose the correct statement



(1) $T_1 > T_2$

(2) $T_2 > T_1$

(3) $T_1 = T_2$

(4) None of these

14. A satellite is orbiting round the earth at a height h above the surface of earth. If this distance h is increased, the time period of satellite will

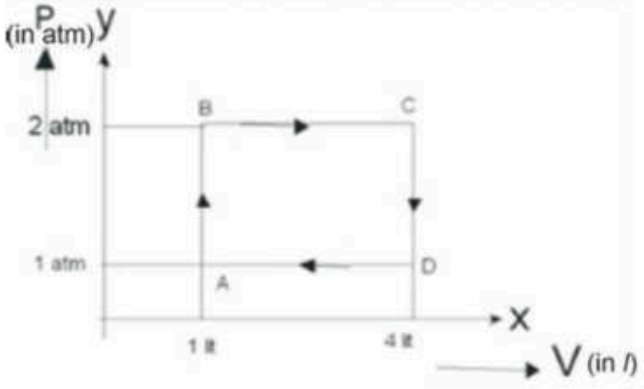
(1) Decrease

(2) Increase

(3) Remain unaffected

(4) Become zero

15. An ideal gas undergoes a cyclic process 'ABCDA' as shown. The net heat absorbed by the gas during the cycle is nearly (assume that One-liter atmosphere is 100 joules).



(1) 600 J

(2) 300 J

(3) 1200 J

(4) 150 J

16. The velocity of liquid coming out of a hole, in a tank placed on ground filled with liquid, in the side wall is

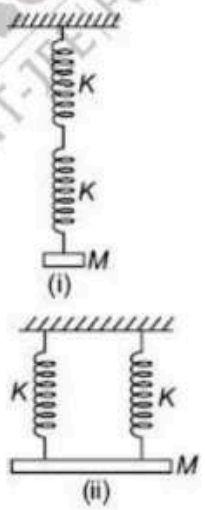
(1) More if hole is at upper end

(2) Same everywhere if liquid of very high density is used

(3) More if hole is at centre of wall

(4) More if hole is near bottom of tank

17. Two identical springs are connected to mass m as shown.



If the time period in (i) case is 4 s. The time period of (ii) case will be

(1) 2 s

(2) 4 s

(3) $2\sqrt{2}$ s

(4) $4\sqrt{2}$ s

18. A geostationary satellite (time period 24 h) is orbiting the earth at height $6R$ above the earth surface where R is radius of earth. The time period of another satellite at a height $\frac{3R}{4}$ from earth surface would be

(1) 24 h

(2) $\frac{6}{2.5}$ h

(3) $\frac{2.5}{6}$ h

(4) 3 h

19. The amplitude of a wave disturbance propagating in positive x direction is given by $y = \frac{1}{1+x^2}$; at time $t = 0$ and $y = \frac{1}{1+(x-1)^2}$; at time $t = 2$ seconds where y and x are in meter. The speed of the wave is

(1) 2 m/s

(2) 0.1 m/s

(3) 0.4 m/s

(4) 0.5 m/s

20. A pulse is generated at lower end of a hanging rope of uniform density and length L. The speed of the pulse when it reached the mid-point of rope is

(1) $\sqrt{2gL}$

(2) $\sqrt{gL}$

(3) $\sqrt{\frac{gL}{2}}$

(4) $\frac{\sqrt{gL}}{2}$

21. A string is divided into three segments, so that the segments have fundamental frequencies in the ratio 1 : 2 : 3. The length of three segments are in the ratio

(1) 4 : 3 : 2

(2) 4 : 2 : 1

(3) 3 : 2 : 1

(4) 6 : 3 : 2

22. If a charge of –1C moves from a point at potential 3 V to another point at potential 2 V

(1) Its potential energy increases

(2) Its potential energy decreases

(3) Its kinetic energy remains unchanged

(4) Its kinetic energy increases

23. If a wire is stretched to become double of its original length, then its resistance

(1) Decreases by 50%

(2) Decreases by 40%

(3) Increases by 300%

(4) Increases by 400%

24. If two identical wires have resistivity ρ_1 and ρ_2. They are connected in series, then the equivalent resistivity is equal to

(1) $\rho_1 \rho_2$

(2) $\rho_1 + \rho_2$

(3) $\frac{\rho_1+\rho_2}{2}$

(4) $\frac{\rho_1-\rho_2}{2}$

25. A wire is connected to a battery and drift velocity of electrons in the conductor is v_0. If wire is stretched to quadruple its length and connected to same battery, the new drift velocity is

(1) $4v_0$

(2) v_0

(3) $\frac{v_0}{4}$

(4) $\frac{v_0}{2}$

26. **Statement(A):** The total charge of electric dipole is obviously zero. This means that the electric field due to electric dipole will be zero.
Statement(B): Direction of electric dipole moment is from negative charge to positive charge.

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both the statements (A) and (B) are correct

(4) Neither statement (A) nor (B) is correct

27. A body has – 48 μC of charge. Number of excess electrons in it will be

(1) 5×10^{14}

(2) 4.8×10^{-6}

(3) 3×10^{14}

(4) 8×10^{-5}

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28. A steel wire of length 4 m and diameter 5 mm is stretched by 5 kg-wt. The increase in its length, if the Young's modulus of steel wire is 2.4×10^{12} dyne cm^{-2}, is

(1) 0.041 cm

(2) 0.0041 cm

(3) 0.00041 cm

(4) 0.41 cm

29. Temperature of a gas is increased from 127°C to 527°C. The rms speed of gas molecules will increase to a factor of

(1) 2

(2) 4

(3) $\sqrt{2}$

(4) $\sqrt{3}$

30. The internal energy of an ideal gas depends upon

(1) Volume

(2) Pressure

(3) Temperature

(4) Density

31. A : Strain is unitless quantity.
R : Strain is equivalent to force.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

32. The temperature of a wire is doubled. The Young's modulus of elasticity will be

(1) Double

(2) Four times

(3) Remain same

(4) Will decrease

33. A body floats in water with 60% of its volume inside water. Same body floats in other liquid with 50% of its volume outside the liquid. The relative density of the liquid is

(1) $\frac{3}{2}$

(2) $\frac{6}{5}$

(3) $\frac{3}{4}$

(4) $\frac{3}{5}$

34. If a simple harmonic motion is represented by, $\frac{d^2x}{dt^2} + \alpha x = 0$, its time period is

(1) $2\pi\sqrt{\alpha}$

(2) $2\pi\alpha$

(3) $\frac{2\pi}{\sqrt{\alpha}}$

(4) $\frac{2\pi}{\alpha}$

35. Liquid *A* and *B* are at 40°C and 20°C when mixed in equal masses. Their specific heat are in ratio of 3 : 2, find the temperature of the mixture

(1) 24°C

(2) 28°C

(3) 32°C

(4) 36°C

SECTION-B

36. The time period of oscillation of a particle whose position varies with time *t* as $x = (\sin\omega t + \sin 2\omega t + \sin 4\omega t)$ is

(1) $\frac{2\pi}{\omega}$

(2) $\frac{\pi}{\omega}$

(3) $\frac{\pi}{2\omega}$

(4) $\frac{\pi}{4\omega}$

37. Two soap bubbles of radii 3 cm and 4 cm come in contact to form a double bubble in air. The radius of curvature of the common interface of the two bubbles is

(1) 3 cm

(2) 4 cm

(3) 12 cm

(4) 23 cm

38. The equation of a plane progressive harmonic wave is given by $y = (5\ \text{cm})\sin(2\pi t + 1.5\pi x)$. The maximum medium particle velocity is

(1) 10 cm/s

(2) 15 cm/s

(3) 10 π cm/s

(4) 20 cm/s

39. Water in river 20 m deep is flowing at a speed of 10 m s^{-1}. The shear stress between horizontal layers of water in river in (N m^{-2}) is (η = coefficient of viscosity of water = 10^{-3} SI unit)

(1) 10^{-2}

(2) 5×10^{-3}

(3) 10^{-3}

(4) 5×10^{-4}

40. The escape velocity of a body on an imaginary planet which has half the radius of earth and double the average density of earth, is (v_e is escape velocity on earth)

(1) $\frac{v_e}{\sqrt{2}}$

(2) $\sqrt{2}\,v_e$

(3) $2\sqrt{2}\,v_e$

(4) $\frac{2}{\sqrt{3}}v_e$

41. Two fixed charged particles 9 μC and 16 μC are separated by a distance 40 cm. The point where net electrostatic field is zero is

(1) 12 cm from 9 μC charge

(2) $\frac{120}{7}$cm from 9 μC charge

(3) 8 cm from 9 μC charge

(4) $\frac{120}{7}$ cm from 16 μC charge

42. A capacitor of capacity C_1 = 10 μF is charged to 40 V & a second capacitor of capacity C_2 = 15 μF is charged to 30 V. If they are connected in parallel, with like charged plates, the charge flow is

(1) 20 μC

(2) 45 μC

(3) 60 μC

(4) 75 μC

43. In the experiment of the meter bridge if length corresponding to null deflection is *x*, what would be its value if the radius of meter bridge wire is doubled?

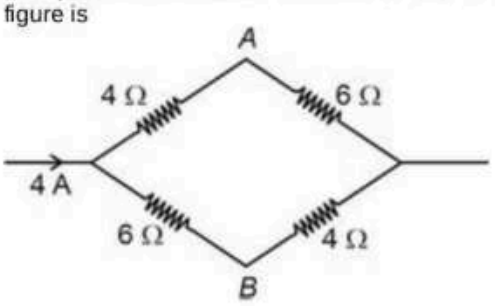
(1) *x*

(2) $\frac{x}{4}$

(3) 4*x*

(4) 2*x*

44. The potential difference between *A* and *B* as shown in figure is



(1) 2 V

(2) 6 V

(3) 4 V

(4) 1 V

45. 100% conversion of heat into work is impossible even when the engine used is ideal. It implies from

(1) First law of thermodynamics

(2) Second law of thermodynamics

(3) Zeroth's law of thermodynamics

(4) Kinetic theory of gases

46. Consider the following two statements:
I. Internal energy of a gas depends on the state of gas.
II. In an isothermal process change in internal energy is maximum.
Then the correct option is

(1) Only I is correct

(2) Only II is correct

(3) Both I and II are correct

(4) Neither I nor II is correct

47. The elastic potential energy stored in a wire is 1J for an extension of 1 mm. The extra energy stored in the wire if the length is further increased by 1 mm is.

(1) 1 J

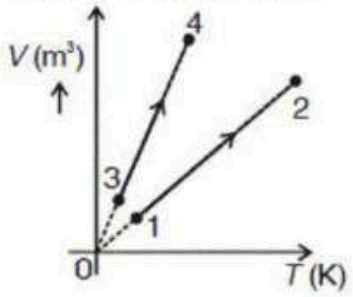
(2) 2 J

(3) 3 J

(4) 4 J

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48. Volume versus temperature ($V - T$) graph of an ideal gas of equal number of moles is plotted as shown in the figure. Choose the correct alternative.



(1) $P_1 = P_2, P_3 = P_4$ and $P_3 < P_1$

(2) $P_1 = P_2, P_3 = P_4$ and $P_4 > P_2$

(3) $P_1 = P_2 = P_3 = P_4$

(4) $P_4 > P_3 > P_1 > P_2$

49. On centigrade scale the temperature of a body increases by 30°C. The increase in temperature on Fahrenheit scale is

(1) 50°F

(2) 40°F

(3) 30°F

(4) 54°F

50. In an open organ pipe, if fundamental frequency is f, then 3rd overtone will be

(1) $4f$

(2) $8f$

(3) $5f$

(4) $7f$

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CHEMISTRY

SECTION-A

51. Conjugate acid and conjugate base of HPO_4^{2-} respectively are

(1) $H_2PO_4^-$ and H_3PO_4

(2) $H_2PO_4^-$ and PO_4^{3-}

(3) H_3PO_4 and PO_4^{3-}

(4) PO_4^{3-} and $H_2PO_4^-$

52. 100 mL of 0.01 M solution of NaOH is diluted to 1 litre. The pH of resultant solution will be

(1) 3

(2) 12

(3) 11

(4) 8

53. If the pH of saturated solution of $M(OH)_3$ is 8 then the value of its solubility product (K_{sp}) at 25°C will be

(1) 3.11×10^{-24}

(2) 8×10^{-30}

(3) 3.33×10^{-25}

(4) 2×10^{-27}

54. The reaction for which, $K_p = K_c$ is satisfied for

I. $A(g) + 2B(g) \rightleftharpoons 3C(g)$
II. $A(s) \rightleftharpoons B(g)$
III. $2A(g) \rightleftharpoons B(g) + C(g)$
IV. $A(s) + B(g) \rightleftharpoons C(s) + 2(s) + 2D(g)$

(1) I and II only

(2) II and III only

(3) I and III only

(4) I, II and IV only

55. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**
Assertion : At equilibrium, concentration of reactants and products become constant.
Reason : A catalyst does not alter the equilibrium constant of a reaction.
In the light of the above statements, choose the **correct** answer from the options given below

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

56. The limiting ionic molar conductivities of K^+, Al^{3+}, SO_4^{2-} ions are x, y and z S cm^2 mol^{-1} respectively. The λ_m^o value for potash alum is

(1) (2x + y + 4z) S cm^2 mol^{-1}

(2) (2x + 2y + 4z) S cm^2 mol^{-1}

(3) (2x + 2y – 4z) S cm^2 mol^{-1}

(4) $\left(\frac{x}{4}+\frac{3y}{4}+z\right)$ S cm^2 mol^{-1}

57. If the standard reduction potential of three redox couples A^{2+}/A, B^{2+}/B and C^{2+}/C are –2.1 V, 1.5 V and –0.5 V respectively then the reducing power of A, B and C will be in order of .

(1) B > A > C

(2) C > A > B

(3) B > C > A

(4) A > C > B

58. The number of moles of $Cr_2O_7^{2-}$ required to react with two moles of sulphite ion in acidic solution is

(1) $\frac{1}{3}$

(2) 1

(3) $\frac{2}{3}$

(4) $\frac{2}{5}$

59. The indicator used to detect the end point of the reaction in iodometric titration is

(1) Methyl red

(2) Phenolphthalein

(3) Starch

(4) Methyl orange

60. The compound in which all the C atoms are sp^2 hybridised is

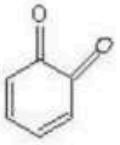
(1) $CH_3 - CH = CH - CH_3$

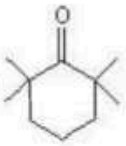
(2) $CH_2 = CH - CH_2 - CH_3$

(3) $CH_2 = CH - CH = CH_2$

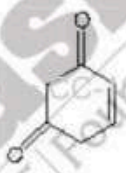
(4) $CH_2 = CH - C \equiv CH$

61. Tautomerism is shown by

(1) 

(2) 

(3) 

(4) 

62. Which of the following statement(s) is/are correct?
A. Hyperconjugation involves delocalisation of σ electrons of C – H bond of an alkyl group directly attached to an atom of unsaturated system.
B. Hyperconjugation involves delocalisation of σ electrons of C – H bond of an alkyl group directly attached to an atom with an unshared *p* orbital.
C. Hyperconjugation is a permanent effect.

(1) A only

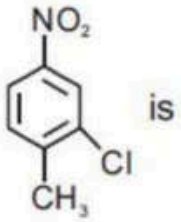
(2) A and B only

(3) A and C only

(4) A, B and C

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63. IUPAC name of the given compound is



(1) 2-Chloro-1-methyl-4-nitrobenzene

(2) 4-Methyl-5-chloro-nitrobenzene

(3) 4-Methyl-3-chloro-1-nitrobenzene

(4) 1-Chloro-2-methyl-5-nitrobenzene

64. Consider the following statements.
(a) There are infinite number of conformations of ethane.
(b) The repulsive interaction between the electron clouds, which affects stability of a conformation, is called torsional strain.
(c) In ethane, magnitude of torsional strain depends upon the angle of rotation about C–C bond.
The **correct** statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

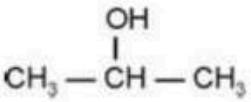
65. In the reaction,

$CH_3–CH=CH_2 \xrightarrow[(ii)\ H_2O_2/\ OH^-]{(i)\ B_2H_6/\ THF} P$ (Major),

Product P is

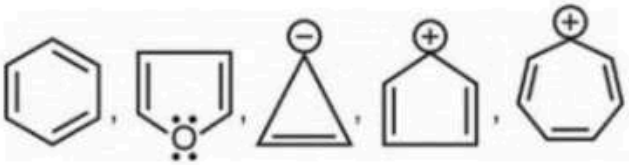
(1) $CH_3CH_2CH_2OH$

(2) $CH_2 = C = CH_2$

(3) 

(4) $HC \equiv C - CH_3$

66. Among the given species



The total number of aromatic species is

(1) 2

(2) 3

(3) 4

(4) 5

67. Given below are the two statements
Statement I: Alkyl fluorides on reduction with zinc and dilute HCl gives alkanes.
Statement II: Wurtz reaction is done in presence of dry ether.
In light of the above statements, choose the **correct** answer.

(1) Statement I is correct but statement II is incorrect.

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

68. Which of the following is not true for solution showing positive deviation from Raoult's law?

(1) $\Delta V_{mix} > 0$

(2) $\Delta H_{mix} > 0$

(3) $\Delta S_{mix} > 0$

(4) $\Delta G_{mix} > 0$

69. If elevation in boiling point for aqueous Na_2SO_4 having α = 40% is ΔT_b then its depression in freezing point would be

(1) $\frac{K_b}{\Delta T_b \times K_f}$

(2) $\frac{1.8 \times K_b}{\Delta T_b \times K_f}$

(3) $\frac{\Delta T_b \times K_f}{1.8 \times K_b}$

(4) $\frac{\Delta T_b \times K_f}{K_b}$

70. Aqueous solution of which of the following has lowest freezing point? ($K_f(H_2O)$ = 1.86 K molal^{-1})

(1) 0.1 m NaCl

(2) 0.01 m NaCl

(3) 0.001 m NaCl

(4) 0.0001 m NaCl

71. Given below are two statements one is assertion(A) other is reason (R).
Assertion (A): Mixture of chloroform and acetone shows positive deviation from Raoult's law.
Reason (R): Chloroform and acetone form intermolecular hydrogen bond.
In the light of above statements, choose the **correct** answer.

(1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.

(2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.

(3) Assertion is correct, but reason is wrong statement.

(4) Assertion is wrong, but reason is correct statement.

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72. The gas used in tanks by scuba divers for breathing contains

(1) 78% N_2, 21% O_2 and 1% He

(2) 11.7% He, 58% N_2 and 30.3% O_2

(3) 32.1% O_2 and 67.1% He

(4) 11.7% He, 56.2% N_2 and 32.1% O_2

73. Electrolyte used in lead storage battery is

(1) 20% solution of NaOH

(2) 38% solution of H_2SO_4

(3) Moist paste of NH_4Cl and $ZnCl_2$

(4) 10% solution of NaCl

74. A solution of $AgNO_3$ is electrolysed for 20 minutes with a current of 9.65 ampere using inert electrode. The mass of silver deposited at cathode is (Atomic mass of silver = 108 u)

(1) 18.25 g

(2) 10.8 g

(3) 12.96 g

(4) 21.6 g

75. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion A: Conductivity always decreases with decrease in concentration for both strong and weak electrolytes.
Reasons R: The number of ions per unit volume that carry the current in a solution decreases on dilution.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

76. $Cu^+ + e^- \rightarrow Cu$, $E^\circ = x_1$ volt

$Cu^{2+} + 2e^- \rightarrow Cu$, $E^\circ = x_2$ volt, then for

$Cu^{2+} + e^- \rightarrow Cu^+$, E°(volt) will be:

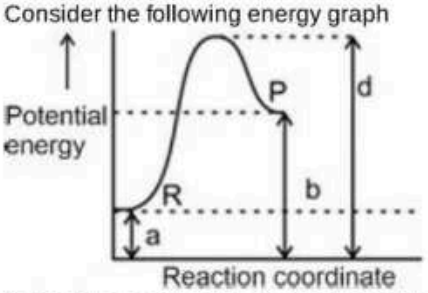
(1) $x_1 - 2x_2$

(2) $x_1 + 2x_2$

(3) $x_1 - x_2$

(4) $2x_2 - x_1$

77. Consider the following energy graph



Activation energy of the forward reaction is

(1) a

(2) a + b

(3) d – a

(4) b – a

78. Which among the following is an incorrect statement?

(1) The reaction taking place in one step is called elementary reaction

(2) Order of a reaction is an experimental quantity

(3) Order as well as molecularity of a reaction can be non integer

(4) For a complex reaction, rate of reaction is given by slowest step of the reaction

79. Mechanism of a hypothetical reaction
$A_2 + B_2 \rightarrow 2AB$
is given below :

(i) $A_2 \rightleftharpoons A + A$ (fast)

(ii) $A + B_2 \rightarrow AB + B$ (slow)

(iii) $A + B \rightarrow AB$ (fast)

The order of reaction with respect to A_2 is

(1) Zero

(2) 0.5

(3) 1.5

(4) 2

80. If rate constant 'k' of a reaction varies with temperature as $\ln k = 2 - \frac{200}{T}$, then Arrhenius constant (A) and activation energy (E_a) of the reaction respectively are

(1) 100, 200 R

(2) 100, 20 R

(3) e^2, 200 R

(4) e^2, 20 R

81. $H_3C-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH=CH_2 + HBr \rightarrow A$

A (predominantly) is

(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-\underset{\underset{Br}{|}}{CH}-CH_3$

(2) $CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{Br}{|}}{CH}-CH_2-CH_3$

(3) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2-CH_2-Br$

(4) $CH_3-\overset{\overset{Br}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_2-CH_3$

82. The hydrocarbon which on monochlorination gives a single monochloride is

(1) $CH_3-CH_2-CH_2-CH_2-CH_3$

(2) $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$

(3) $CH_3-\overset{\overset{CH_3}{|}}{CH}-CH_2-CH_3$

(4) $CH_3-CH_2-CH_2-CH_3$

83. $CHCl_3 + O_2 \xrightarrow{light} \underset{(major)}{P}$

The product P is

(1) CO_2

(2) CO

(3) $COCl_2$

(4) CCl_4

84. Least reactive towards S_N1 reaction is

(1) CH_3Cl

(2) $CH_2 = CH - Cl$

(3) $CH_2 = CH - CH_2Cl$

(4) CH_3CH_2Cl

85. 2-Bromo-3-methylpentane is heated with potassium ethoxide in ethanol. The major product obtained is

(1) 3-Methylpent-2-ene

(2) 2-Ethoxy-3-methylpentane

(3) 2-Methylpent-2-ene

(4) 4-Methylpent-2-ene

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SECTION-B

86. 50% of a first order reaction was completed in 6.93 min, then 75% of the same reaction would be completed in

(1) 2.303 min

(2) 13.86 min

(3) 69.09 min

(4) 23.03 min

87. If for a first order reaction $t_{90\%}$ is 50 s then the value of $t_{99.9\%}$ will be

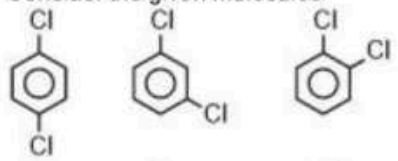
(1) 150 s

(2) 100 s

(3) 10 s

(4) 50 s

88. Consider the given molecules



(I) (II) (III)

The correct order of melting point will be

(1) I > III > II

(2) III > II > I

(3) II > III > I

(4) I > II > III

89. Consider the following sequence of reactions

$$C_2H_5C \equiv CH \xrightarrow{NaNH_2} A \xrightarrow{CH_3CH_2Br} B.$$

The product (B) is

(1) $C_2H_5CH = C = CH_2$

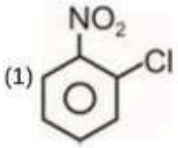
(2) $C_2H_5C \equiv C - CH_2CH_3$

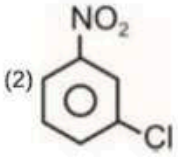
(3) $C_2H_5 - \underset{\displaystyle NH_2}{\overset{\displaystyle |}{C}} = CH - CH_2 - CH_3$

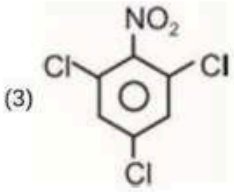
(4) $C_2H_5C = CH_2$ with CH_2CH_3 as a branch

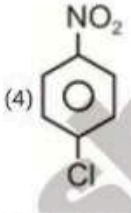
90. $C_2H_2 \xrightarrow[Fe\ \ tube]{Red\ \ hot} A \xrightarrow[Conc.\ HNO_3]{Conc.\ H_2SO_4} B \xrightarrow[Anhyd.\ AlCl_3]{Cl_2} C\ (Major)$

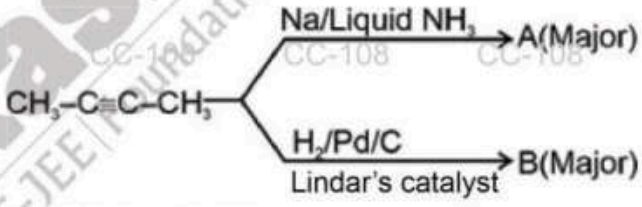
The product C in the above reaction sequence is

(1) 

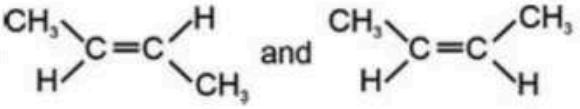
(2) 

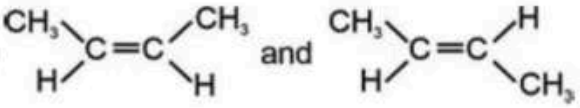
(3) 

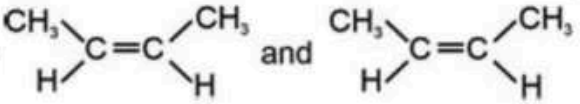
(4) 

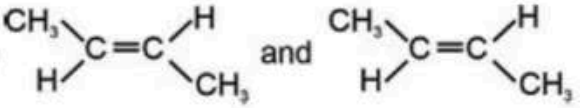
91. 

A and B respectively are

(1) 

(2) 

(3) 

(4) 

92. Total number of structural isomers of C_4H_8 is

(1) 4

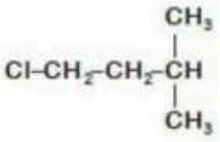
(2) 5

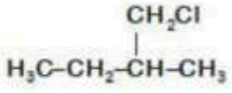
(3) 6

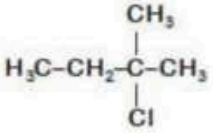
(4) 3

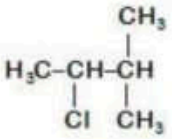
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93. An alkene "A" on reaction with O_3 and $Zn–H_2O$ gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is:

(1) 

(2) 

(3) 

(4) 

94. The pH of 10^{-8} M NaOH solution is

(1) 6.98

(2) 6

(3) 7.02

(4) 8

95. Which of the following reactions is an example of metal displacement reaction?

(1) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

(2) $TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$

(3) $3Mg + N_2 \rightarrow Mg_3N_2$

(4) $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$

96. In which of the following cases pH is less than 7? [pK_a (CH_3COOH) = 4.74, pK_b (NH_4OH) = 4.74]

(1) 50 ml of 0.1 M CH_3COOH + 50 ml of 0.1 M KOH

(2) 50 ml of 0.1 M HCl + 50 ml of 0.1 M NaOH

(3) 50 ml of 0.1 M HNO_3 + 50 ml of 0.1 M NH_4OH

(4) 50 ml of 0.1 M H_2SO_4 + 50 ml of 0.2 M NaOH

97. If the equivalent conductance of $\frac{M}{20}$ solution of a weak monobasic acid is 25.0 S cm^2 eq^{-1} and at infinite dilution is 250 S cm^2 eq^{-1} then the dissociation constant of the acid is

(1) 2.5×10^{-3}

(2) 5.55×10^{-4}

(3) 2.5×10^{-5}

(4) 5.55×10^{-5}

98. If O_2 (g) is bubbled through water at 293 K. Calculate the millimoles of O_2(g) which would dissolve in 10 mol of water at oxygen gas pressure of 0.5 atm ($K_H(O_2)$ at 293 K is 3.5×10^4 atm).

(1) 0.143

(2) 0.234

(3) 1.2

(4) 14.3

99. Three Faradays of electricity were passed through $AgNO_3$(l), $CuSO_4$(l) and $AlCl_3$(l) kept in three different vessels using Pt electrodes. The ratio of the moles of Ag, Cu and Al deposited will be

(1) 2 : 3 : 6

(2) 6 : 3 : 2

(3) 1 : 2 : 3

(4) 3 : 2 : 1

100. For an aqueous solution of $K_4[Fe(CN)_6]$ (strong electrolyte) the van't Hoff factor (i) will be

(1) 4

(2) 1

(3) 3

(4) 5

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BOTANY

SECTION-A

101. Mesophyll tissue is differentiated into palisade and spongy parenchyma in

(1) Grass leaf

(2) Monocot leaf

(3) Isobilateral leaf

(4) Dicot leaf

102. The gritty texture in guava fruit is due to the abundance of

(1) Sclereids

(2) Suberin deposition in fruit wall

(3) Sclerenchyma fibres

(4) Parenchyma tissues

103. As compared to monocot root, the dicot root has/ shows

(1) Sclerenchymatous pericycle

(2) Fewer xylem bundles

(3) Large and well developed pith

(4) Primary growth only

104. The function of root hair and trichome respectively are

(1) Absorption of water along with minerals and gaseous exchange

(2) Transpiration and gaseous exchange

(3) Gaseous exchange and protection

(4) Absorption of minerals and preventing transpiration

105. **Assertion (A):** The plant factors affecting photosynthesis are dependent on the genetic predisposition of the plant.
Reason (R): At any point, the rate of photosynthesis will be determined by the factor available at sub-optimal levels.
Consider the above statements and choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

106. **Statement A:** Protonema is formed in mosses and liverworts.
Statement B: Protonema represents sporophytic stage of bryophytes.
Read the above statements and select the **correct** option.

(1) Only (A) is correct

(2) Only (B) is correct

(3) Both (A) and (B) are incorrect

(4) Both (A) and (B) are correct

107. Select the **mismatched** pair.

(1) Lycopsida – *Selaginella*

(2) Sphenopsida – *Equisetum*

(3) Pteropsida – *Adiantum*

(4) Psilopsida – *Dryopteris*

108. Agar is found in the cell wall of

(1) Red algae

(2) Brown algae

(3) Green algae

(4) Blue green algae

109. The event precursor to the seed habit is seen in

(1) Liverworts

(2) Pteridophytes

(3) Algae

(4) Mosses

110. Cytotaxonomy is based on all of the following, **except**

(1) Chromosome number

(2) DNA sequencing

(3) Chromosome structure

(4) Chromosome behaviour

111. Select the **incorrect** statement w.r.t. bryophytes.

(1) The plant body is attached to the substratum by unicellular or multicellular rhizoids

(2) They may possess stem-like and leaf-like structures

(3) The sporophyte is always free-living and photosynthetic

(4) Zygote does not undergo reduction division immediately

112. Which of the given is **not** a C_4 plant?

(1) *Sorghum*

(2) Maize

(3) *Amaranthus*

(4) Tomato

113. How many ATP and NADPH molecules are required to fix six molecules of CO_2 through Calvin cycle?

(1) 6 ATP and 6 NADPH

(2) 3 ATP and 2 NADPH

(3) 18 ATP and 12 NADPH

(4) 12 ATP and 12 NADPH

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114. Read the following statements and choose the **correct** option.
(a) In T.W. Engelmann experiment, the bacteria were used to detect the sites of O_2 evolution.
(b) The chloroplasts align themselves along the walls of the mesophyll cells, such that they get the optimum quantity of the incident light.
(c) Accessory pigments protect only chlorophyll *c* from photo-oxidation.
(d) Cyclic photophosphorylation can occur when light of wavelengths only beyond 680 nm are available for excitation.

(1) Only (a) and (c) are correct

(2) Only (b) and (d) are correct

(3) Only (d) and (c) are correct

(4) Only (a), (b) and (d) are correct

115. A major limiting factor which influences the rate of photosynthesis is

(1) CO_2 concentration

(2) Light quality

(3) Duration of light

(4) Water availability

116. The pigment that appear blue-green in the chromatogram is

(1) Chlorophyll *a*

(2) Chlorophyll *b*

(3) Carotene

(4) Xanthophyll

117. Complex that is involved in ETS and has two copper centres is

(1) NADH dehydrogenase

(2) Succinate dehydrogenase

(3) Cytochrome *c* oxidase complex

(4) Cytochrome bc_1 complex

118. When fatty acid tripalmitin is used as respiratory substrate, the RQ will be

(1) 1

(2) 0.7

(3) 0.9

(4) 1.33

119. Select the **incorrect** statement about glycolysis.

(1) It takes place in all cellular living organisms

(2) It forms a 2C molecule as end product

(3) It takes place in cytoplasm of cell

(4) It consumes ATP molecules at a few steps

120. In glycolysis, which of the given steps involves ATP utilisation?

(1) Glucose-6-phosphate → Fructose-6-phosphate

(2) Glyceraldehyde-3-phosphate → 1,3-bisphosphoglyceric acid

(3) Fructose-6-phosphate → Fructose 1,6-bisphosphate

(4) 1,3-bisphosphoglyceric acid → 3-phosphoglyceric acid

121. Opening and closing of stomata is controlled by

(1) Guard cells

(2) Subsidiary cells

(3) Bulliform cells

(4) Mesophyll cells

122. Which range of wavelength constitute the PAR?

(1) Light between 430-550 nm

(2) Light between 400-700 nm

(3) Light between 700-780 nm

(4) Light between 680-700 nm

123. Identify the algae A and B and select the **correct** option.



(1) A - *Fucus*, B - *Porphyra*

(2) A - *Laminaria*, B - *Chara*

(3) A - *Dictyota*, B - *Porphyra*

(4) A - *Dictyota*, B - *Polysiphonia*

124. Guard cells of stomata

(1) Lack chloroplasts

(2) Have thick outer wall in dicots

(3) Are dumb bell shaped in grasses

(4) Surround subsidiary cells

125. The scheme of glycolysis was given by

(1) Gustav Embden, Otto Meyerhof, and J. Parnas

(2) Hans Krebs and Parnas

(3) Calvin and Hans Krebs

(4) Calvin and Otto Meyerhof

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126. The most abundant enzyme in the world is

(1) RuBisCO

(2) NADP reductase

(3) PEPcase

(4) ATP synthase

127. Which of the following scientists isolated the auxin from tips of coleoptiles of oat seedlings?

(1) Charles Darwin

(2) F. Skoog

(3) Francis Darwin

(4) F.W. Went

128. A hormone which promotes female flowers in cucumber is

(1) Gibberellin

(2) Cytokinin

(3) Auxin

(4) Ethylene

129. Formation of tracheary elements in primary xylem is an example of

(1) Differentiation

(2) Dedifferentiation

(3) Redifferentiation

(4) Fragmentation

130. Select the **incorrectly** matched pair.

(1)	Gibberellins	–	Promotes bolting
(2)	Auxins	–	Promote apical dominance
(3)	Ethylene	–	Initiates sprouting of potato tuber
(4)	2, 4-D	–	Used to kill mature monocot weeds

(1) (1)

(2) (2)

(3) (3)

(4) (4)

131. Shedding of pollen grains in 60% of angiosperms occurs at A celled stage.

Select the **correct** option for 'A'.

(1) One

(2) Two

(3) Three

(4) Four

132. How many meiotic divisions are required to produce 480 seeds?

(1) 479

(2) 600

(3) 70

(4) 650

133. Which of the following options about tapetum is **correct**?

(1) It is a protective layer of thick walled cells

(2) It is a sporogenous layer

(3) It nourishes the developing pollen grains

(4) It helps in dehiscence of anther

134. The another name of ovule is

(1) Megaspore

(2) Megasporangium

(3) Male gametophyte

(4) Embryo sac

135. Inner thin layer of the cell wall of pollen grain is made up of

(1) Cellulose and pectin

(2) Sporopollenin only

(3) Most resistant organic material

(4) Lipids and sporopollenin

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SECTION-B

136. Match the structures given in column-I with their explanation/meaning given in column-II and select the **correct** option.

Column I	Column II
a. Coleoptile	(i) Seedless
b. Perisperm	(ii) Hollow foliar structure
c. Parthenocarpic fruits	(iii) Undifferentiated sheath
d. Coleorrhiza	(iv) Residual persistent nucellus

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(ii), b(iii), c(iv), d(i)

137. Aleurone layer is a part of

(1) Endosperm

(2) Embryo

(3) Testa

(4) Tegmen

138. Pollen grains of cereals, such as wheat, lose viability _____ of their release.
Select the **correct** option to fill in the blank.

(1) After several months

(2) In 20 hours

(3) After 60 minutes

(4) Within 30 minutes

139. Vascular bundle is conjoint and open in

(1) Dicot root

(2) Monocot root

(3) Dicot stem

(4) Monocot stem

140. Lateral roots arise from

(1) Epidermis

(2) Pericycle

(3) Hypodermis

(4) Pith

141. Read the following statements and select the **correct** option.
Statement A: Phloem fibres are made up of sclerenchymatous cells.
Statement B: Bast fibres are generally absent in secondary phloem.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are correct

(4) Both statements are incorrect

142. Products of light reaction are

(1) O_2 and CO_2

(2) CO_2 and H_2O

(3) ATP and NADPH

(4) ATP and CO_2

143. Plants with Kranz anatomy do **not** have/show

(1) Double fixation of CO_2

(2) Calvin cycle

(3) 4C acids in their mesophyll cells

(4) Oxygenase activity of RuBisCO

144. Select the **mis-matched pair.**

(1) F_0–F_1 particles – Inner mitochondrial membrane

(2) End product of glycolysis – Acetyl CoA

(3) TCA cycle – Elucidated by Hans Krebs

(4) Molecular oxygen – Electron acceptor in ETS

145. In mitochondria, when electrons move through ETS, as a result protons accumulate in the

(1) Matrix

(2) Outer membrane

(3) Perimitochondrial space

(4) Inner membrane

146. The F_1 head piece in mitochondria is a

(1) Catalytic site which produces ADP from ATP

(2) Channel through which protons cross the outer membrane

(3) Integral membrane protein complex

(4) Peripheral membrane protein complex

147. Gibberellins are responsible for all of the given, **except**

(1) Bolting

(2) Induce seed dormancy

(3) Increase malt yield from barley grains

(4) Delay fruit ripening

148. _____ is called antitranspirant hormone as it stimulates the closure of stomata.

(1) Auxin

(2) GA

(3) ABA

(4) Cytokinin

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149. The cells proximal to the meristematic zone of root are characterised by

(1) Increased vacuolation and cell enlargement

(2) Wall thickening and protoplasmic modification

(3) Dense protoplasm and large nuclei

(4) Attaining maximum size but lack nucleus

150. Leaves of conifers do **not** have

(1) Needle like shape

(2) Very thin cuticle

(3) Sunken stomata

(4) Ability to withstand extremes of temperature

ZOOLOGY

SECTION-A

151. Chief secretory constituent of sebaceous glands is/are

(1) Sterols

(2) Urea

(3) Glucose

(4) Amino acids

152. The cortex extends in between the medullary pyramids as renal columns called

(1) Hilum

(2) Calyces

(3) Columns of Bertini

(4) Capsule

153. The part of renal tubule which is impermeable to water is

(1) Descending limb of loop of Henle

(2) Ascending limb of loop of Henle

(3) PCT

(4) Collecting duct

154. Read the following statements w.r.t nitrogenous wastes.
Statement A : Ammonia is most toxic and requires the large amount of water for its elimination.
Statement B : Uric acid is least toxic and can be removed with a minimum loss of water from the body.
Choose the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement B is incorrect

(4) Only statement A is incorrect

155. Dialysing fluid after the process of hemodialysis additionally contains

(1) Albumin

(2) Nitrogenous wastes

(3) Globulin

(4) Fibrin

156. Acromion process articulates with the

(1) Head of femur

(2) Head of humerus

(3) Collar bone

(4) Sternum

157. White muscle fibres differ from red muscle fibres in

(1) Having high content of myoglobin

(2) Containing plenty of mitochondria

(3) Having reddish appearance

(4) Having high amount of sarcoplasmic reticulum

158. Which of the following is considered as anatomical unit of muscle?

(1) Sarcomere

(2) Fasciculus

(3) Muscle fibre

(4) Fascia

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159. The type of muscle fibres present in our

(1) Biceps are uninucleated and voluntary

(2) Heart are non-striated and involuntary

(3) Thigh are multinucleated and fusiform in shape

(4) Stomach are unbranched and involuntary

160. What is the total number of cranial and facial bones in an adult human?

(1) 14

(2) 22

(3) 80

(4) 126

161. Which one is **not** the part of brain stem?

(1) Forebrain

(2) Midbrain

(3) Pons

(4) Medulla oblongata

162. Resting membrane potential is maintained by the sodium-potassium pump which transports

(1) 2 Na^+ outwards for 2 K^+ into the cell

(2) 3 K^+ outwards for 3 Na^+ into the cell

(3) 3 Na^+ outwards for 2 K^+ into the cell

(4) 3 Na^+ outwards for 3 K^+ into the cell

163. Which part of human brain is concerned with regulation of sexual behavior and expression of emotional reactions?

(1) Mid brain

(2) Medulla oblongata

(3) Limbic lobe along with hypothalamus

(4) Cerebellum

164. The gastric secretion and cardiovascular reflexes are controlled by centres located in

(1) Medulla oblongata

(2) Cerebellum

(3) Pons

(4) Mid brain

165. During resting stage, the axonal membrane is comparatively more permeable to **A** and nearly impermeable to **B**.
Select the option that fills the blank **correctly**.

(1) A = Sodium ions, B = Potassium ions

(2) A = Potassium ions, B = Sodium ions

(3) A = Calcium ions, B = Sodium ions

(4) A = Calcium ions, B = Potassium ions

166. Excessive loss of water in urine is prevented mainly by the action of

(1) ADH

(2) ANF

(3) Androgens

(4) Adrenaline

167. The 24 hour diurnal rhythm of our body such as the sleep-wake cycle is regulated by the hormone

(1) Adrenaline

(2) Melatonin

(3) Calcitonin

(4) Prolactin

168. Gonadotropin releasing hormone and gonadotropins are respectively the products of

(1) Posterior pituitary and hypothalamus

(2) Hypothalamus and posterior pituitary

(3) Hypothalamus and anterior pituitary

(4) Anterior pituitary and hypothalamus

169. Select the pair of hormones synthesised by the hypothalamus and transported axonally to the pituitary gland.

(1) Prolactin and TSH

(2) Oxytocin and vasopressin

(3) FSH and LH

(4) MSH and prolactin

170. Excessive bone resorption will take place due to hypersecretion of

(1) PTH

(2) TCT

(3) LH

(4) ACTH

171. A hormone which regulates the growth of mammary glands and milk production in them, is released by

(1) Pars distalis

(2) Pars intermedia

(3) Hypothalamus

(4) Pars nervosa

172. The hormone which acts on both pancreas and gall bladder and stimulates the secretion of pancreatic enzymes and bile juice respectively is

(1) CCK

(2) GIP

(3) Gastrin

(4) Secretin

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173. Which one is **not** a characteristic of members of the phylum Arthropoda?

(1) Metamerism

(2) Jointed appendages

(3) Closed circulation

(4) Triploblasty

174. The phylum in which triploblasty appeared for the first time is

(1) Ctenophora

(2) Aschelminthes

(3) Annelida

(4) Platyhelminthes

175. Which of the following characters are applicable to ctenophores?

(1) Organ system level of organisation

(2) Reproduction asexually only

(3) Direct development

(4) Eight external rows of ciliated comb plates

176. Which of the following animals does not shed their scales as skin cast?

(1) *Naja*

(2) *Calotes*

(3) *Balaenopiera*

(4) *Vipera*

177. In _____, notochord is present only in larval tail, while in _____, it extends from head to tail region.
Select the option that fills the blanks **correctly**.

(1) Urochordata, Cephalochordata

(2) Hemichordata, Urochordata

(3) Tunicata, Hemichordata

(4) Urochordata, Cyclostomata

178. Select the set of animals which are homeotherms.

(1) *Chelone* and *Naja*

(2) *Corvus* and *Macropus*

(3) *Pteropus* and *Calotes*

(4) *Balaenoptera* and *Chameleon*

179. Read the following statements.
Statement I : In each segment of cockroach, exoskeleton shows the presence of hardened plates called sclerites
Statement II : Sclerites are joined to each other by flexible arthrodial membrane.
Select the **correct** option.

(1) Both statements I and II are incorrect

(2) Both statements I and II are correct

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

180. In cockroach, which structure has an outer layer of thick circular muscles and thick inner cuticle forming six highly chitinous plate called teeth?

(1) Hepatic caeca

(2) Proventriculus

(3) Malpighian tubule

(4) Salivary gland

181. Read the following statements and choose the **correct** option.
Statement A : In cockroach, vision is known as mosaic vision with less sensitivity but more resolution
Statement B : In male cockroach, a pair of spermatheca is present in the 6th segment which opens into the genital chamber

(1) Both statements are incorrect

(2) Both statements are correct

(3) Statement A is correct but B is incorrect

(4) Statement B is correct but A is incorrect

182. In male frog, urinary bladder opens into

(1) Rectum

(2) Cloaca

(3) Ureter

(4) Urinogenital duct

183. Which of the following is not associated with forebrain of frog?

(1) Olfactory lobes

(2) Optic lobes

(3) Cerebral hemispheres

(4) Diencephalon

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184. Number of cranial nerves in a frog are

(1) 16

(2) 20

(3) 24

(4) 28

185. Fertilization and development in frogs is

(1) External and direct

(2) Internal and direct

(3) External and indirect

(4) Internal and indirect

SECTION-B

186. Inner area of the cerebral hemispheres forms white matter due to concentration of

(1) Myelinated tracts

(2) Cell bodies

(3) Nissl's granules

(4) Ganglia

187. Choose the **incorrect** statement w.r.t. electrical synapse.

(1) The membranes of pre-and post-synaptic neurons are in very close proximity.

(2) Transmission of impulses across this synapse is very similar to impulse conduction along a single axon.

(3) Electrical current can flow directly from one neuron into other across these synapses.

(4) Electrical synapses are very common in our body.

188. In an adult human, vertebrae of which categories are fused?

(1) Sacral and coccygeal

(2) Cervical and thoracic

(3) Thoracic and lumbar

(4) Cervical and coccygeal

189. Type of synovial joint present between carpals of an adult human is

(1) Gliding joint

(2) Saddle joint

(3) Hinge joint

(4) Pivot joint

190. Nearly all of the essential nutrients and 70 – 80 per cent of electrolytes and water are reabsorbed in which segment of nephron?

(1) PCT

(2) Ascending limb of loop of Henle

(3) Descending limb of loop of Henle

(4) DCT

191. The specialised cells called podocytes are found in which part of a nephron?

(1) PCT

(2) DCT

(3) Bowman's capsule

(4) Collecting duct

192. Two hormones 'A' and 'B' interact with intracellular receptors and regulate the gene expression. 'A' is steroidal and 'B' is non-steroidal in nature. Select the option that correctly identifies 'A' and 'B' respectively.

(1) Aldosterone, Progesterone

(2) Progesterone, Growth hormone

(3) Testosterone, Thyroxine

(4) Thyroxine, Estrogen

193. Metagenesis is commonly found in the members of which phylum?

(1) Porifera

(2) Cnidaria

(3) Ctenophora

(4) Echinodermata

194. Which of the following organisms possesses water vascular system?

(1) *Asterias*

(2) *Sycon*

(3) *Anopheles*

(4) *Hydra*

195. Air bladder regulates buoyancy in

(1) *Pristis*

(2) *Myxine*

(3) *Pterophyllum*

(4) *Hyla*

196. Long bones are hollow with air cavities is the characteristic feature of

(1) Reptilia

(2) Amphibia

(3) Pisces

(4) Aves

197. Excretion in cockroach is performed by

(1) Malpighian tubules

(2) Feather-like gills

(3) Nephridia

(4) Cloaca

198. On an average, female cockroach produce __A__ oothecae, each containing __B__ eggs.
Choose the option that **correctly** fill the blanks A and B respectively.

(1) 9 – 10, 14–16

(2) 14–16, 9 – 10

(3) 9 – 10, 18 – 20

(4) 14 – 16, 4 – 6

199. Which of the following are absent in the body of frog?

(1) Head, Neck and Abdomen

(2) Head, Neck and trunk

(3) Neck and tail

(4) Head and Neck

200. In frog, the final digestion of food takes place in

(1) Stomach

(2) Cloaca

(3) Intestine

(4) Rectum

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